

Explanation of My Stack

Tools used and why:

1. **Tally.so — Lead Capture Form**
Chosen Tally because it has native webhook support on the free tier, so every time someone submits the form, it automatically fires a POST request to n8n without extra configuration. Initially I tried Google Forms, but it brought unnecessary complexity to trigger webhooks.
2. **n8n Cloud — Automation Engine**
n8n is basically the backbone of the entire system. I picked it over platforms like Zapier mainly because of pricing and flexibility. Instead of charging per action or node, n8n charges per workflow execution, which makes larger workflows much more affordable. The visual node editor also makes the logic easy to build, demonstrate and explain.
3. **Groq (llama 3.3 70B Versatile) — AI Qualification & Writing Mails**
Chosen over OpenAI because Groq's free tier is genuinely generous and extremely fast (sub-second responses). The model handles lead qualification, urgency scoring, sentiment analysis, and email generation all in one call.
4. **Gmail — Lead Communication**
Simplest email solution to connect in n8n with OAuth. Used for the initial auto-response and all follow-up emails.
5. **Telegram Bot — Internal Notifications**
Chosen over Slack because setting up a Telegram bot takes 5 minutes with zero cost and no workspace setup required. Delivers real-time urgent lead alerts to the sales team instantly.
6. **Google Sheets — Lead Tracking & Shared Memory**
Google Sheets acts like a lightweight CRM for the entire workflow. Along with storing lead data, it also works as a shared communication layer between different automations. One workflow writes new lead information, another updates appointment statuses from Calendly, and follow-up workflows read from the same sheet before deciding whether an email should still be sent. It ended up being a very simple but effective way to coordinate everything.
7. **Calendly — Appointment Booking**
Free tier allows creating a bookable consultation link that embeds directly into emails. A webhook integration with n8n ensures that when a lead books, their status in Google Sheets updates automatically to "Appointment Booked", stopping further follow-ups.

Limitations:

1. **n8n free tier concurrency** — The free Cloud plan runs workflow executions sequentially, meaning if two leads submit forms simultaneously, the second one queues behind the first. In production this would be solved with n8n's queue mode on a paid plan.

(I initially tried implementing the entire follow-up system using Wait nodes inside a single workflow, which would have been a much cleaner architecture. However, because the free tier processes executions sequentially, long-running workflows with wait states would block newer executions and quickly become impractical.

To work around this limitation, I separated the follow-ups into independent scheduled workflows that periodically check lead status from Google Sheets before sending emails. In a production setup, enabling n8n's queue mode on a paid plan would allow workflows to run concurrently, making the single-workflow + wait-node approach fully feasible and more scalable.)

2. **No real SMS** — The system currently relies entirely on email communication because adding SMS support would require services like Twilio, which needs a paid account and a registered phone number. Email was used as the primary channel instead.
3. **Follow-up timing is approximate** — The follow-up workflows run on a 15-minute schedule, so a "1 hour follow-up" could arrive anywhere between 1 hour and 1 hour 15 minutes after submission. Acceptable for this use case but not precise.
4. **Groq rate limits** — Although Groq provides extremely fast AI responses on its free tier, it still comes with request limits. If the system were handling higher lead volumes, such as more than 40–60 leads per day, those limits could eventually become a bottleneck. In production, a paid Claude or OpenAI API key would handle this more reliably.
5. **Calendly webhook limitation** — The free Calendly plan has limited webhook support. In production, GoHighLevel or HubSpot would handle booking and CRM updates natively in one platform.
6. **No real memory/context** — Each lead interaction is stateless. The system doesn't remember previous conversations with the same lead. In a production-ready version, conversation history would typically be stored in a database and passed back into the AI model so it can generate more personalized and context-aware responses over time.